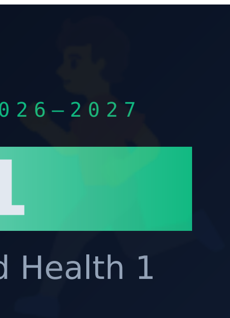
PATHFIT 1 · QCU BSIT

Theory Notes

Written-Exam Content Only

QCU ACADEMIC OS · AY 2026–2027



PATHFIT 1

Physical Activities Toward Health 1 — Theory Notes

Course: PATHFIT 1 | Units: 2 units

Type: Mostly Practical/Activity-Based

OVERVIEW

Why This Guide Is Shorter

Most of your PATHFIT 1 grade comes from actually performing — fitness assessments, attendance, activity participation. No study guide can substitute for training. This guide covers ONLY the written/theory slice that typically shows up on paper exams: fitness concepts, terminology, and the principles behind why your instructor structures workouts the way they do.

How to Actually Use This

Read this once to understand the vocabulary your instructor will use in class (so instructions like "focus on muscular endurance today" or "we're testing your flexibility" make sense), then focus your real effort on showing up and performing consistently — that's where your grade actually lives.

CHAPTER 01

Physical Fitness Components

Physical fitness splits into two categories: components that matter for general HEALTH, and components that matter for athletic SKILL/performance. Your instructor will reference both, and paper exams commonly ask you to classify or define these.

Health-Related Fitness Components

Cardiovascular Endurance
Heart/lungs' ability to sustain prolonged activity

Muscular Strength
Max force a muscle can produce in one effort

Muscular Endurance
Ability to sustain repeated muscle effort over time

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Flexibility

Range of motion available at a joint

Body Composition

Ratio of fat mass to lean mass (muscle, bone, organs)

Skill-Related 🏃 Fitness Components

Agility

Ability to change direction quickly and efficiently

Balance

Maintaining body equilibrium, still or moving

Coordination

Using senses and body parts together smoothly

Power

Strength applied quickly (force × speed)

Reaction Time

Time between a stimulus and your response

Speed

Ability to move the body quickly across distance

Exam-Critical Distinction

Health-related components matter for EVERYONE's general wellbeing, regardless of sport. **Skill-related** components matter more for athletic PERFORMANCE in specific sports/activities. If a question asks "which is NOT health-related," the answer is always one of the six skill-related ones (agility, balance, coordination, power, reaction time, speed).

CHAPTER 02

Exercise Principles

These principles explain WHY workouts are structured the way they are — the theory behind your instructor's programming decisions. Frequently tested as identification or short-answer questions.

The FITT Principle

The core framework for designing any exercise program — four variables you adjust to control workout difficulty and results.

Letter	Stands For	Meaning
F	Frequency	How often you exercise (e.g. 3x per week)
I	Intensity	How hard you exercise (e.g. % of max heart rate, weight lifted)
T	Time	How long each session lasts (duration)
T	Type	What kind of exercise (cardio, strength, flexibility)

Other Core Training Principles

Principle	Meaning
Overload	To improve fitness, the body must be challenged BEYOND its current capacity — doing only what's already comfortable produces no adaptation
Progression	Overload must increase gradually over time as the body adapts — jumping too fast risks injury

Principle	Meaning
Specificity	Training adaptations are specific to the type of exercise performed — running builds running endurance, not necessarily swimming endurance
Reversibility	Fitness gains are lost over time if training stops — often summarized as "use it or lose it"
Individual Differences	People respond differently to identical training based on genetics, age, fitness history — one program doesn't fit everyone equally

Quick Memory Anchor

FITT = the DIAL you adjust each session. Overload/Progression/Specificity/Reversibility/Individual Differences = the RULES explaining why adjusting that dial correctly actually produces results over time.

CHAPTER 03

Fitness Testing & Assessment

Standard tests used to measure the components from Chapter 01. Your instructor will likely run several of these in person — knowing what each one actually measures helps you understand why it's being administered.

Test	Measures	How It Works
3-Minute Step Test	Cardiovascular endurance	Step up/down on a platform at a set pace for 3 minutes; recovery heart rate afterward indicates cardio fitness
Push-Up Test	Muscular strength/ endurance (upper body)	Max push-ups performed with correct form, often within a time limit
Curl-Up / Sit-Up Test	Muscular endurance (core)	Max controlled curl-ups performed within a set time
Sit-and-Reach Test	Flexibility (hamstrings, lower back)	Reaching forward from a seated position, measuring how far past the feet you can reach
BMI (Body	Body composition	Weight (kg) ÷ height (m)^2, compared

Test	Measures	How It Works
Mass Index)	(rough estimate)	against standard ranges — a screening tool, not a precise body-fat measurement
Illinois Agility Test	Agility	Timed run through a cone course requiring rapid direction changes

Common Misconception — BMI Isn't Perfect

BMI doesn't distinguish muscle mass from fat mass — a very muscular person can register as "overweight" by BMI despite having low body fat. Instructors and exams often ask you to name this exact limitation.

CHAPTER 04

Exercise Safety & Health Concepts

The practical safety knowledge that keeps a workout effective without causing injury — also common written-exam material.

Warm-Up & Cool-Down

Phase	Purpose
Warm-Up	Gradually raises heart rate and body temperature, increases blood flow to muscles, reduces injury risk before intense activity
Cool-Down	Gradually lowers heart rate, helps prevent blood pooling in the legs (which can cause dizziness), begins the recovery process

Target Heart Rate Zone

A simple, commonly taught formula for estimating maximum heart rate:

Formula	Calculation
Estimated Max Heart Rate	`220 − age`
Moderate-intensity target zone	50–70% of max heart rate
Vigorous-intensity target zone	70–85% of max heart rate

Example: for an 18-year-old, estimated max heart rate = 220 − 18 = **202 bpm**. Moderate zone ≈ 101–141 bpm.

RPE — Rate of Perceived Exertion

A subjective scale (commonly 1–10 or 6–20 depending on the version taught) where you rate how hard an activity FEELS, used alongside or instead of heart rate monitoring — useful since it requires no equipment.

Basic Injury Prevention

- **Proper form** matters more than speed or weight — bad form under load is the most common cause of exercise injury
- **Progressive overload** (Ch. 02) — sudden jumps in intensity/volume are a leading injury cause
- **Hydration** — even mild dehydration measurably reduces performance and increases injury/heat-illness risk
- **Rest and recovery** — muscles adapt and strengthen during rest, not just during the workout itself

Bottom Line for This Course

This theory covers maybe 10–20% of your actual grade. Show up, participate honestly in the fitness assessments, and this vocabulary will make sense of what you're doing and why — that combination is what actually gets you through PATHFIT 1.